

- 7 Great lakes and crater lakes can be found in Nicaragua, a country in Central America. These two types of lake differ in their species richness.

(a) State what is meant by the term **species richness**.

(1)

*(b) Lake Nicaragua and Lake Managua are great lakes.

Crater lakes were formed from the craters of volcanoes.

The photographs show a great lake and a crater lake in Nicaragua.



Great lake

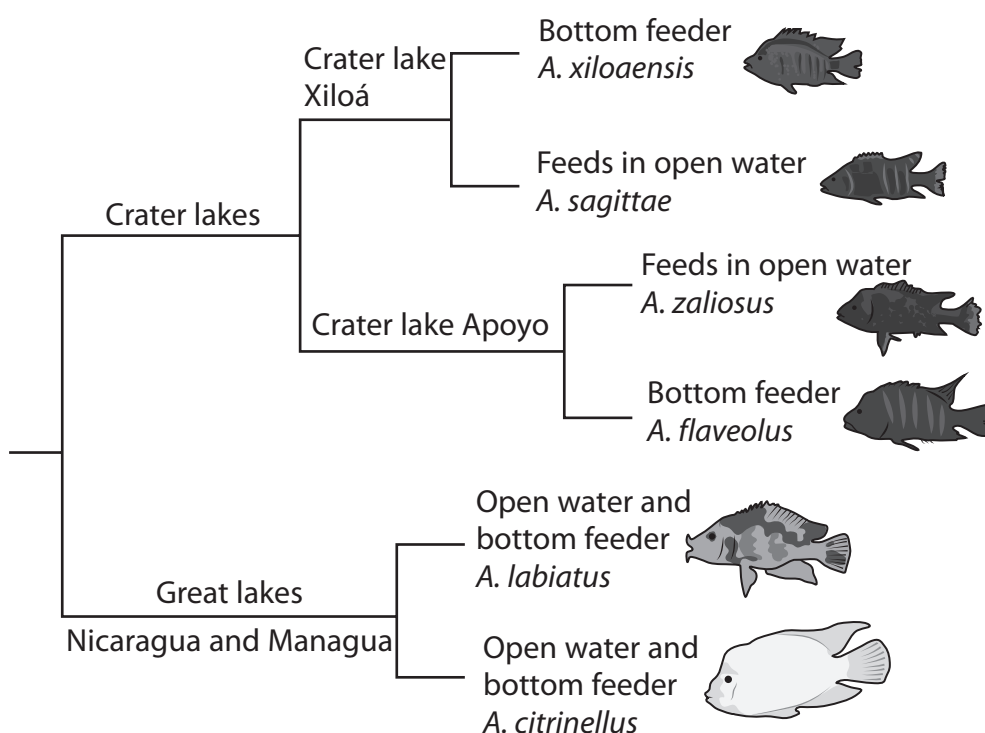
(Source: © mauritius images GmbH/Alamy Stock Photo)



Crater lake

(Source: © John Mitchell/Alamy Stock Photo)

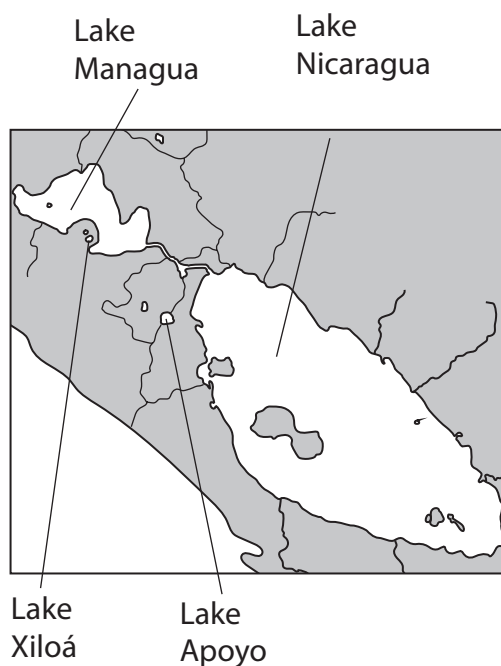
The diagram shows a proposed relationship for some species of cichlid fish and the location of the lakes they inhabit.



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The table shows the estimates of when cichlid fish species first inhabited the crater lakes and when speciation occurred.

Crater lake	First inhabited / number of years ago	First speciation occurred / number of years ago
Apoyo	4700	3700
Xiloá	4300	2700

Describe how new cichlid species may have been formed in these four lakes and how the scientists would have determined that these cichlids were six different species.

Use the information in the question and your own knowledge to support your answer.

(6)

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(c) A study compared the diversity of cichlid species in two crater lakes.

The index of diversity (D) for Lake Xiloá was 2.8.

The table shows the data obtained from Lake Apoyo.

Species	Number of individuals (n)	n(n – 1)
<i>A. astorquii</i>	156	24 180
<i>A. chanco</i>	45	1 980
<i>A. flaveolus</i>	78	6 006
<i>A. globosus</i>	8	56
<i>A. superciliosus</i>	17	272
<i>A. zaliosus</i>	12	
	N = 316	$\sum n(n - 1) =$

An index of diversity (D) is calculated using the formula:

$$D = \frac{N(N - 1)}{\sum n(n - 1)}$$

Determine which of the lakes has the higher biodiversity of cichlids.

Use the table and the formula to help you.

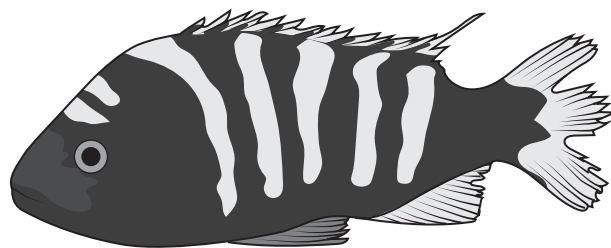
(4)

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(d) The drawing shows two *A. citrinellus* cichlids.



melanic phenotype –
black with white stripes



gold phenotype –
orange with black mottling

The gold phenotype is caused by inheriting a dominant allele and the melanic phenotype is caused by two recessive alleles at the same locus.

A scientist studied the change in allele frequencies in the *A. citrinellus* population over time.

In the first study, the allele frequencies in the population were $p = 0.6$ and $q = 0.4$.

In a later study, 672 fish of this species were gold and 128 were melanic.

Determine if the allele frequencies have changed, using the equations:

$$p^2 + 2pq + q^2 = 1$$

$$p + q = 1$$

(3)

(Total for Question 7 = 14 marks)

